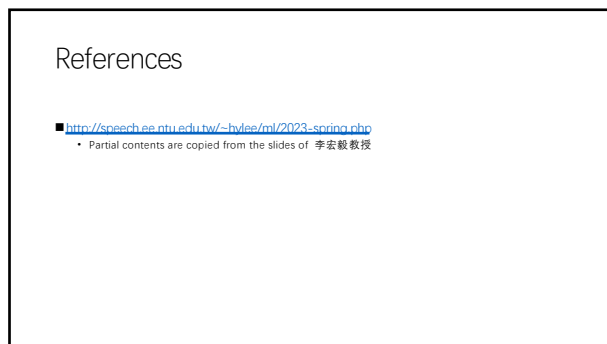




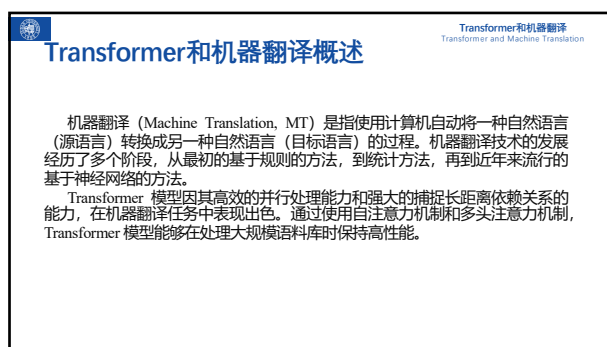
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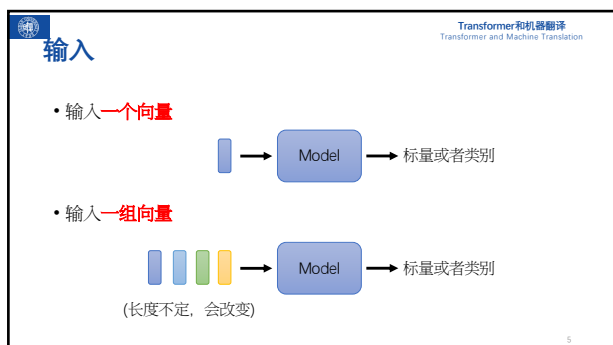
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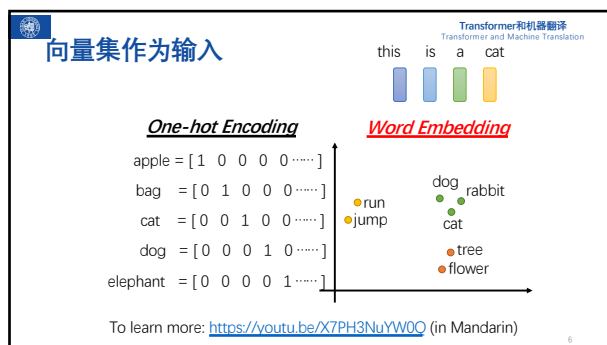
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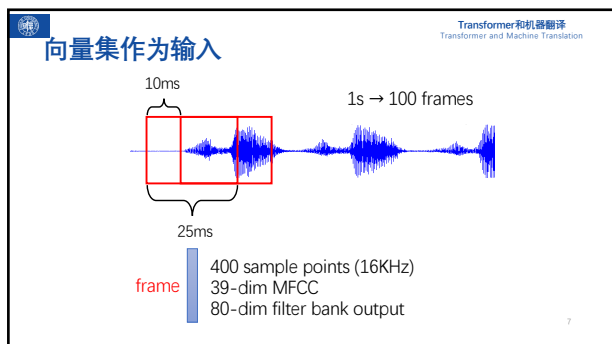
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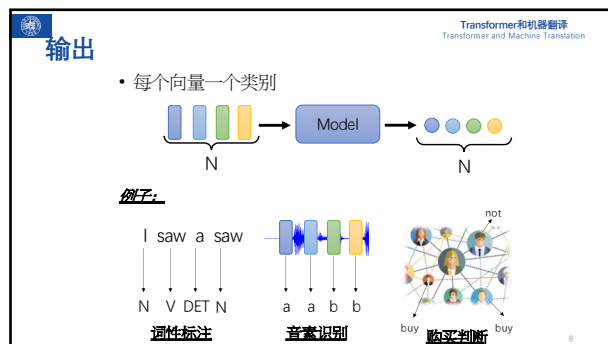
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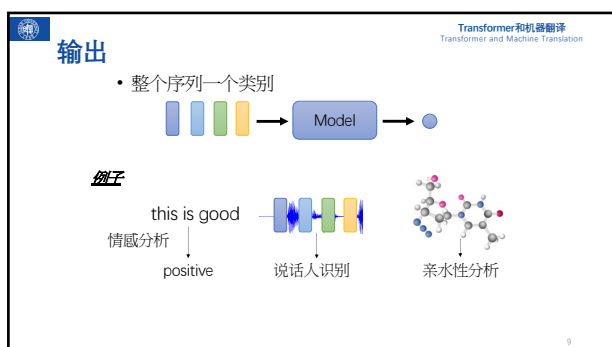
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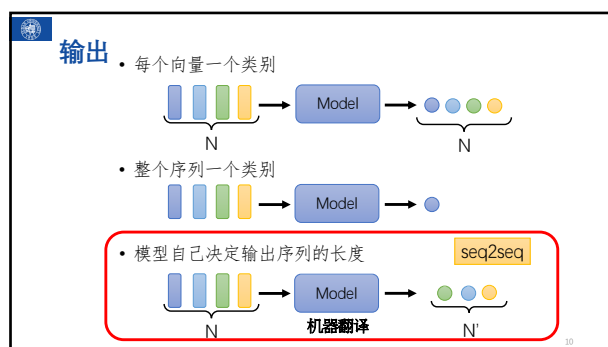
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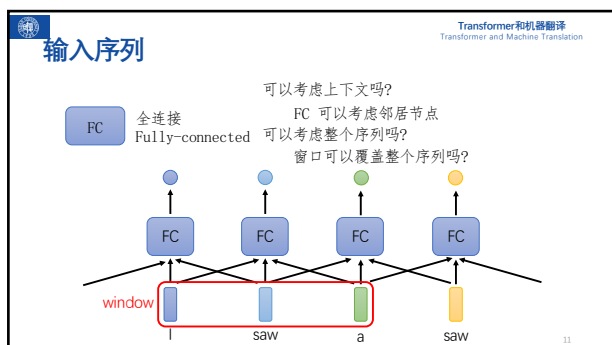
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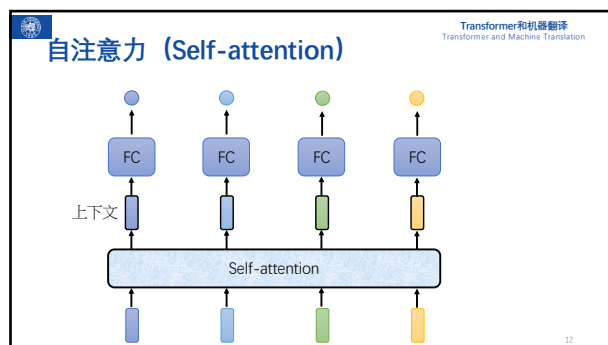
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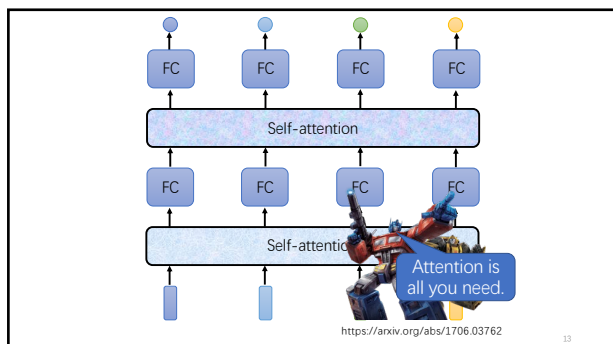
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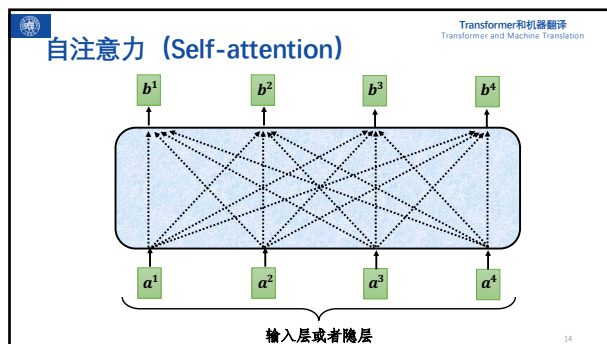
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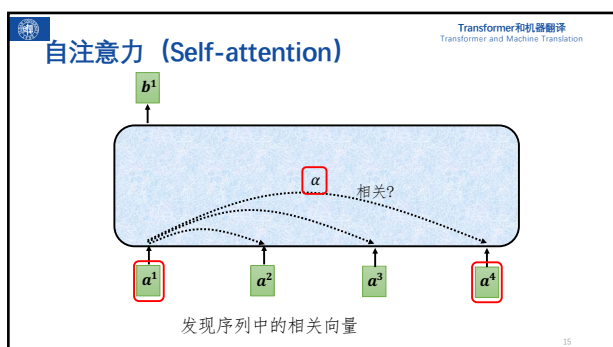
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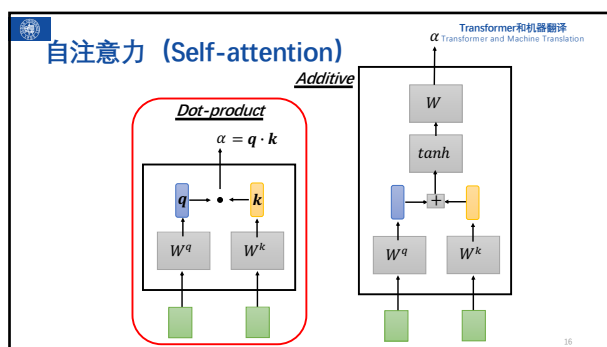
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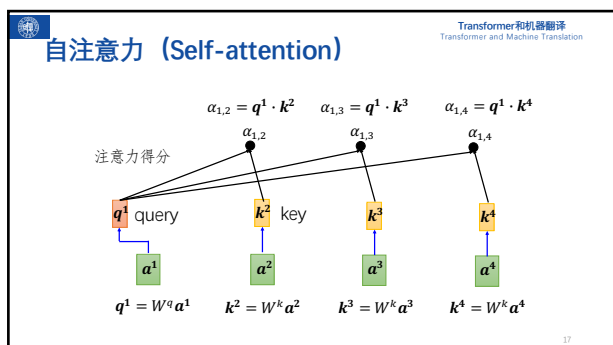
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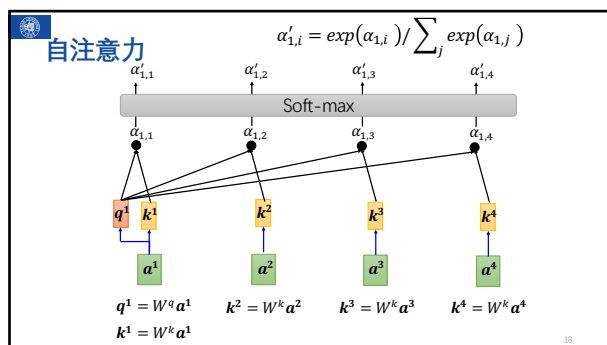
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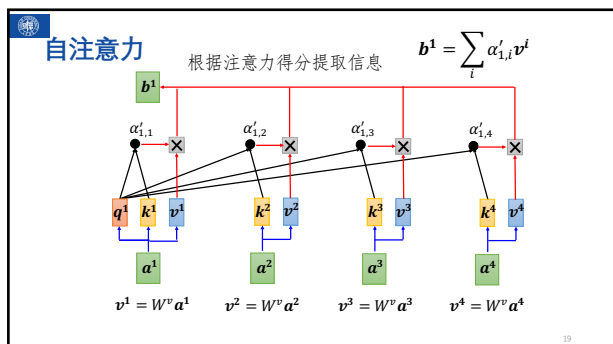
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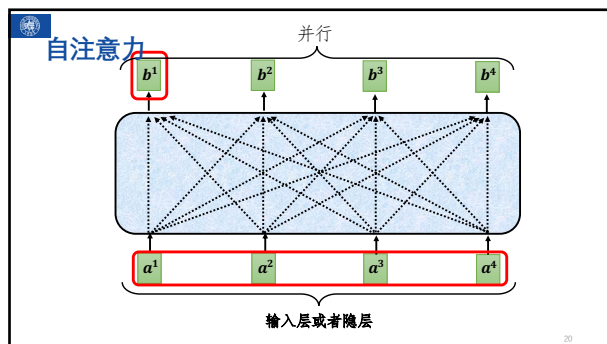
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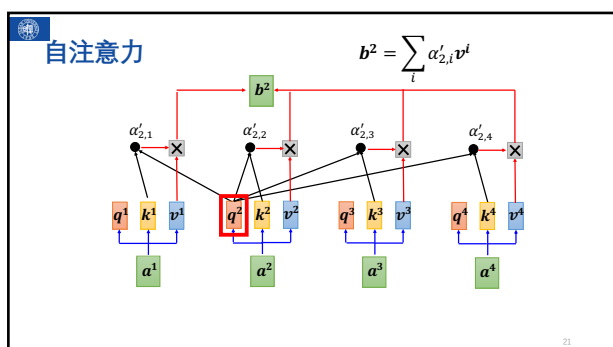
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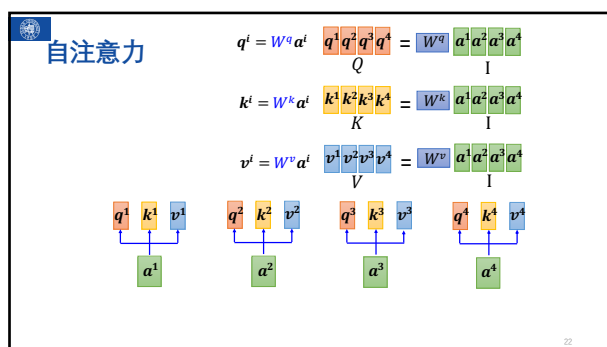
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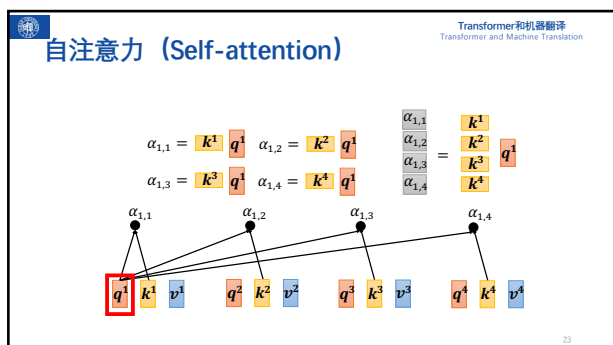
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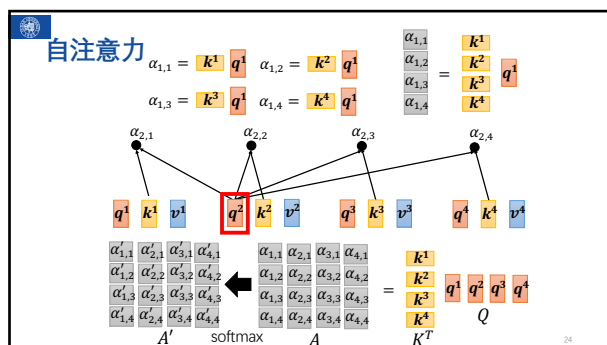
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位置编码

- 自注意力中没有位置信息。
- 每个位置有唯一的位置向量 e^t
- hand-crafted
- learned from data

q^t k^t v^t

$e^t + a^t$

每列表示一个位置向量 e^t

-1 1

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很多应用...

Transformer
<https://arxiv.org/abs/1706.03762>

BERT
<https://arxiv.org/abs/1810.04805>

自注意力广泛用于自然语言处理
(Natural Language Processing, NLP) !

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语音处理中的自注意力

语音是非常长的向量序列

输入序列的长度为 L

注意力矩阵 A'

10ms

关注部分范围

b^1, b^2, b^3, b^4

a^1, a^2, a^3, a^4

Truncated Self-attention

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图像处理中的自注意力

图像也可以看成向量集合

向量

Source of image: https://www.researchgate.net/figure/Color-image-representation-and-RGB-matrix_fig15_282798184

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自注意力和CNN

CNN: 只在感受野中关注的自注意力

➢ CNN 是简化的自注意力

自注意力: 具有可学习感受野的CNN

➢ 自注意力是复杂版的CNN.

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自注意力和CNN

自注意力

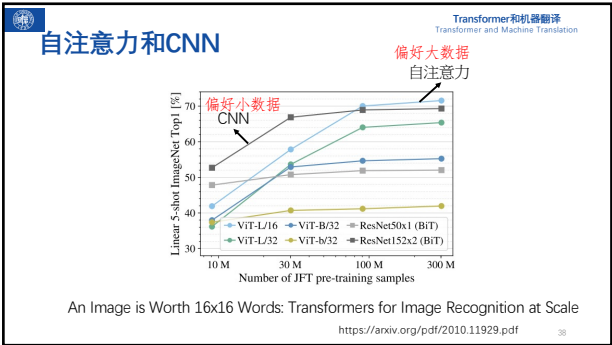
CNN

自注意力与卷积层的关系

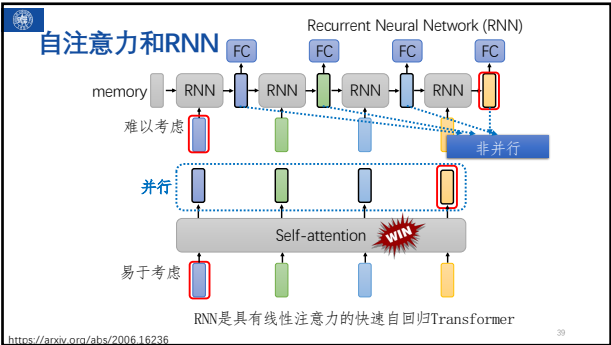
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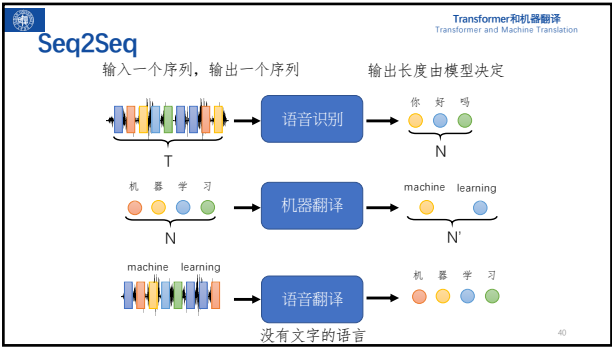
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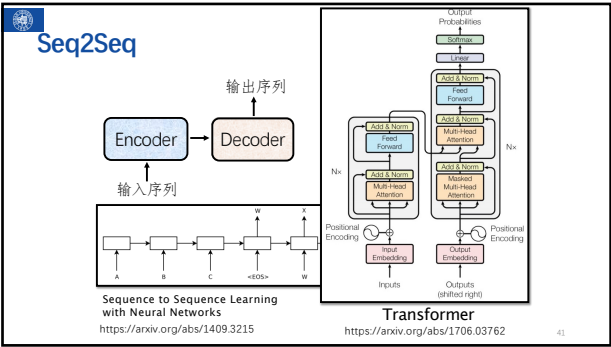
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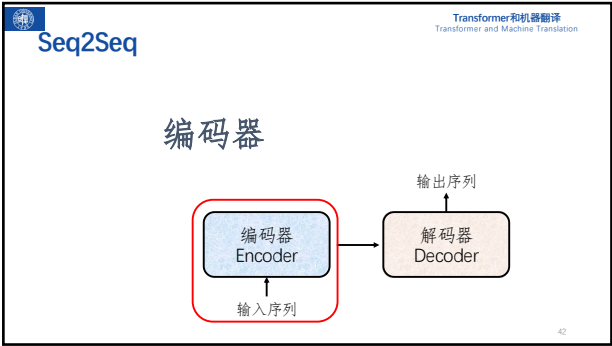
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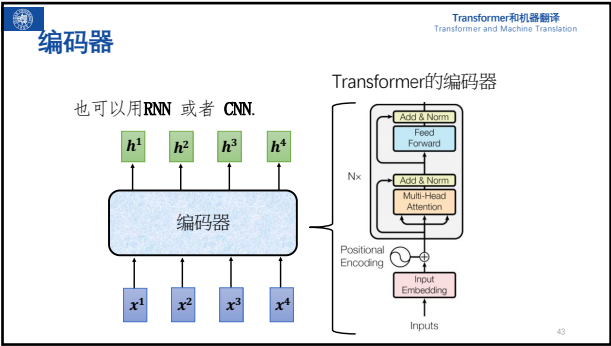
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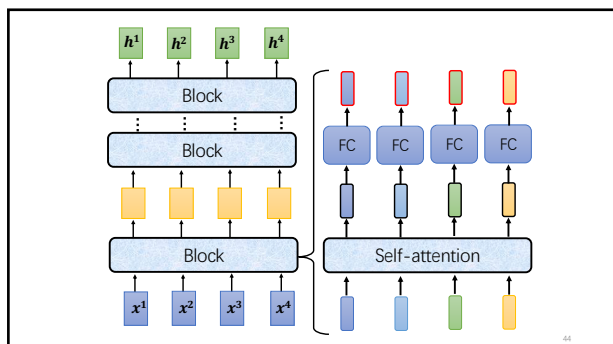
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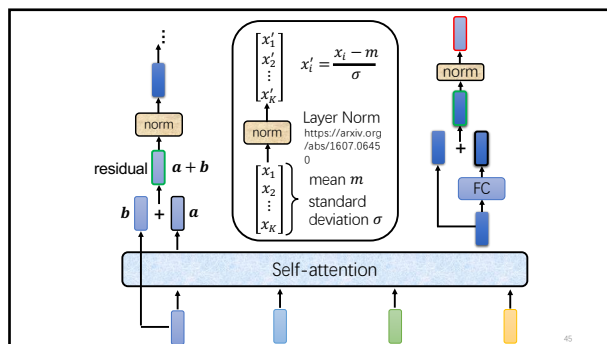
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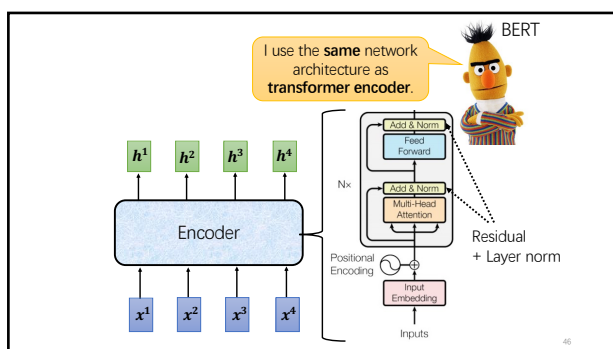
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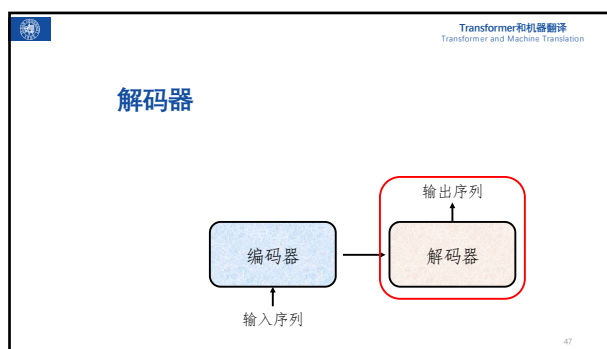
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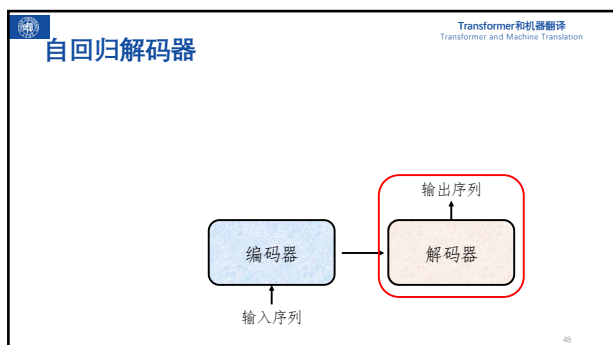
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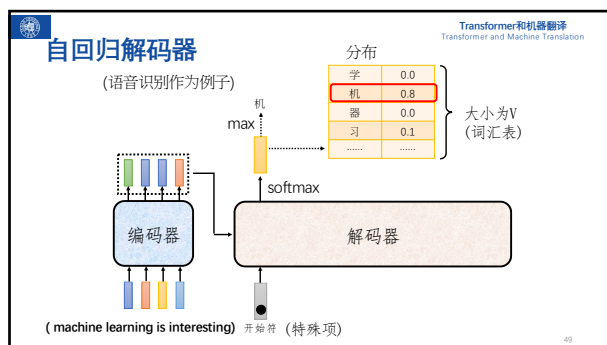
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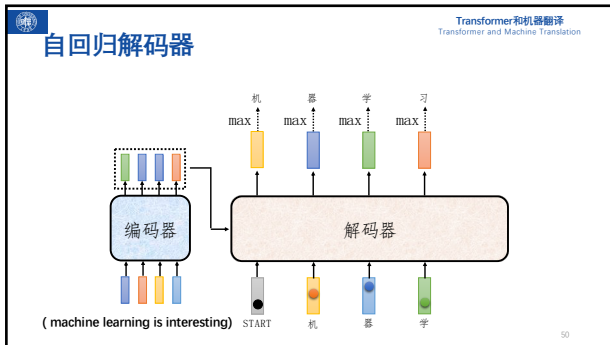
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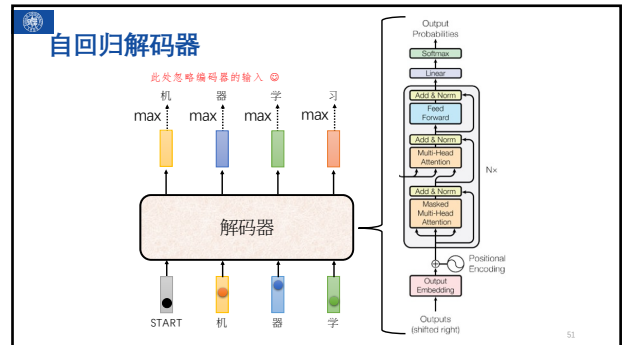
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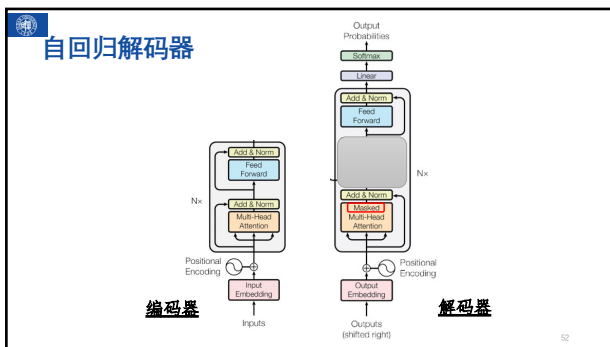
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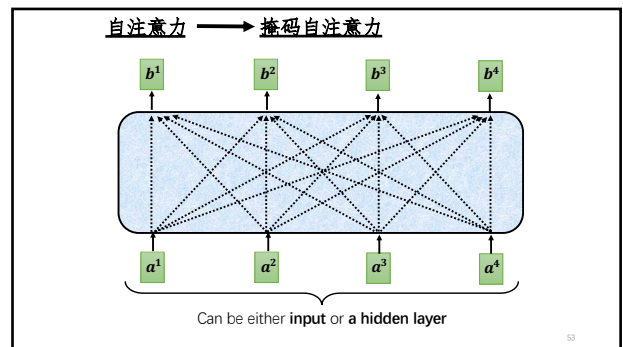
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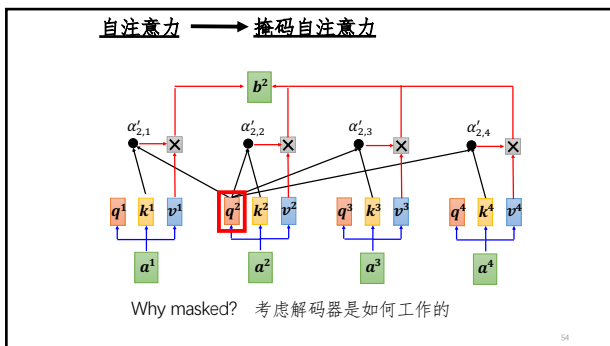
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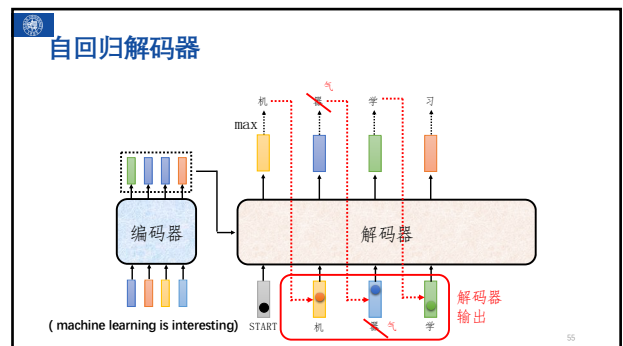
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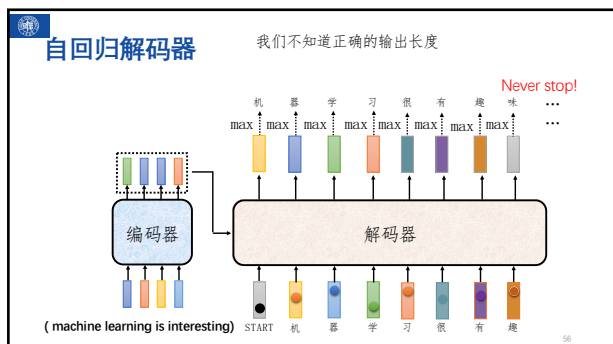
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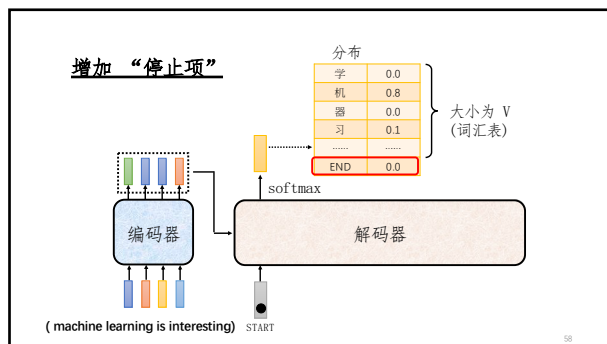
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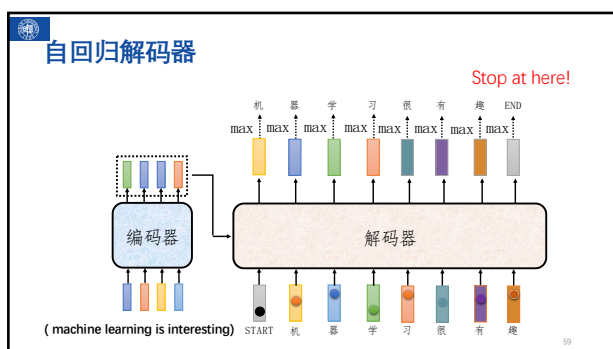
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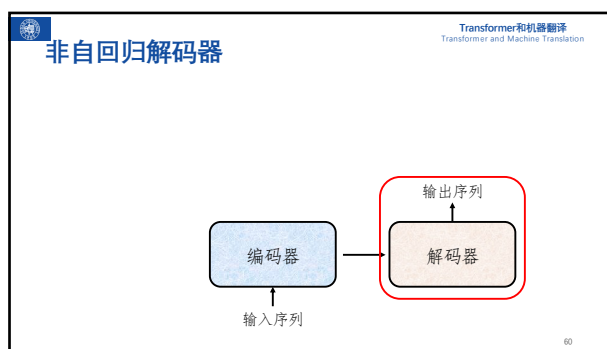
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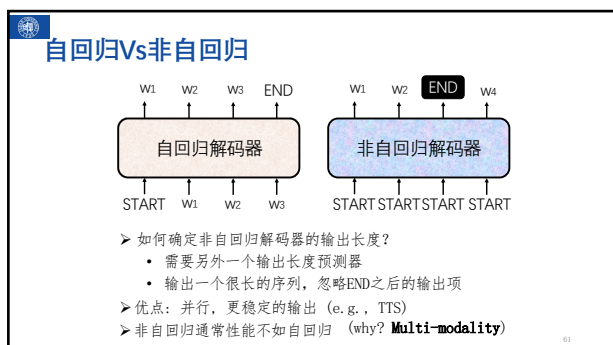
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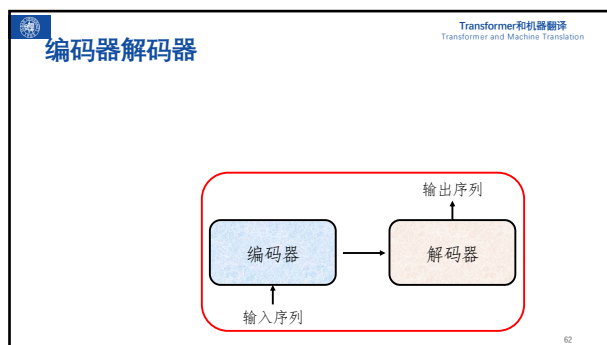
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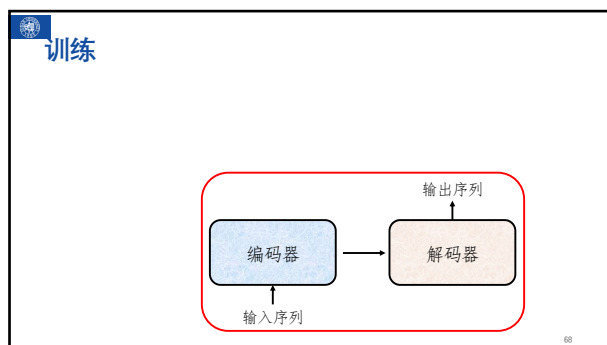
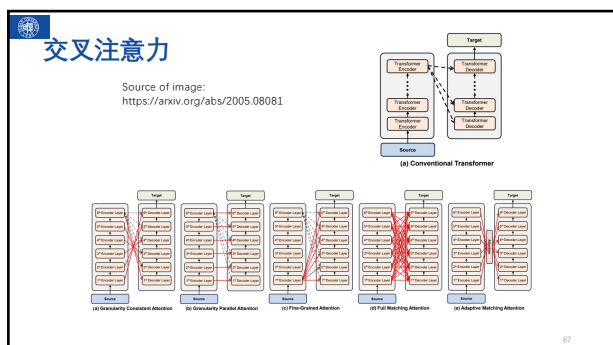
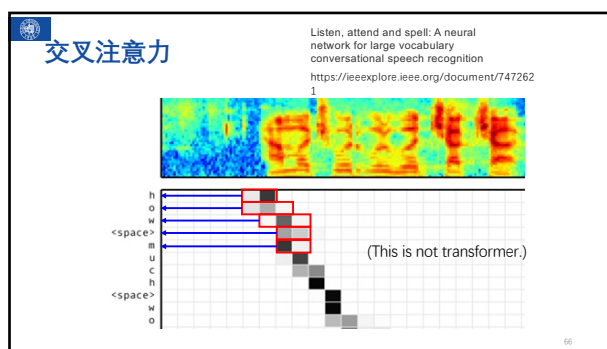
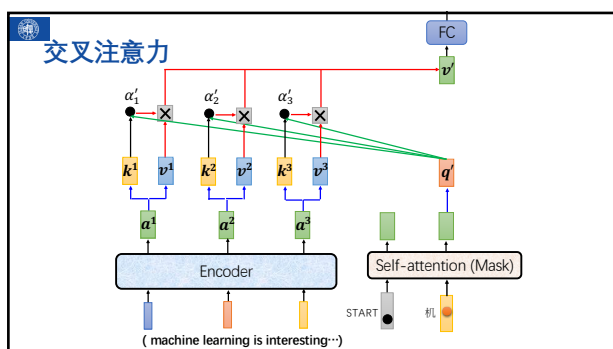
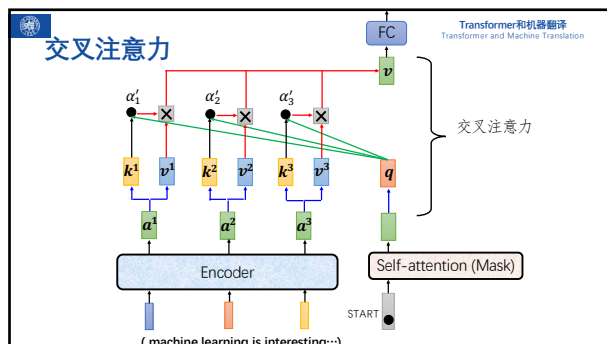
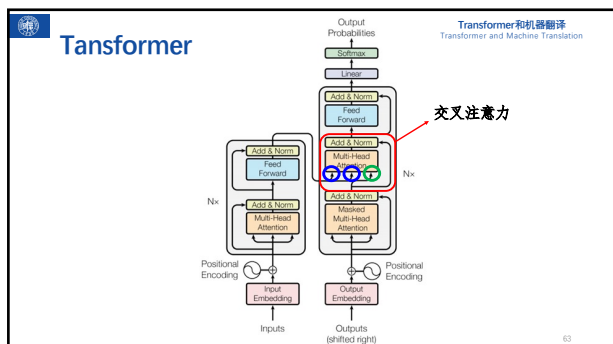
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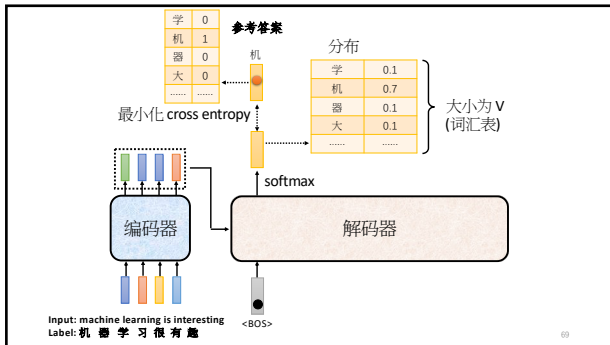


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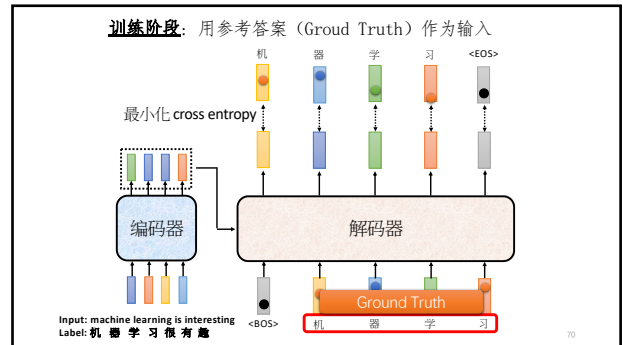


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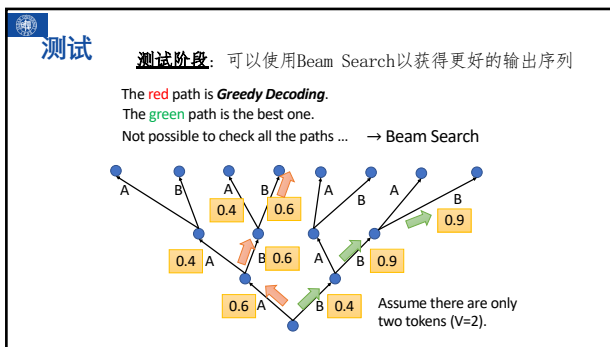




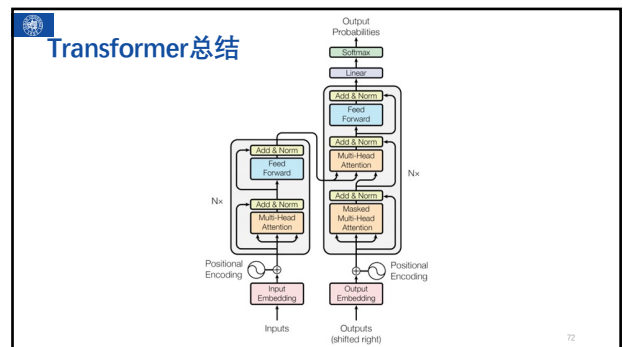
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机器翻译经典模型

Transformer和机器翻译
Transformer and Machine Translation

基于规则的模型

SYSTRAN 最初由美国政府资助, 在20世纪60年代开始研发, 最初的目标是翻译俄语到英语, 以满足情报和军事需求。

Eurotra 是一项由欧洲经济共同体 (EEC) 资助的多语言机器翻译系统研究项目, 始于1978年。该项目的目标是开发一套能够处理欧洲共同体成员国官方语言的机器翻译系统。

统计机器翻译 (SMT) 模型

IBM模型是一系列经典的统计机器翻译模型, 由IBM的研究员们在20世纪80年代末和90年代初提出。这些模型主要用于解决词对齐问题, 即确定源语言句子中的词与目标语言句子中的词之间的对应关系。IBM模型为后来的统计机器翻译系统 (SMT) 奠定了基础, 并且在词对齐算法中有着重要的应用。

神经机器翻译 (NMT)

Transformer模型是由Vaswani等人在2017年的论文《Attention is All You Need》中提出的, 它是一种用于自然语言处理 (NLP) 任务的神经网络架构。Transformer摒弃了传统的递归神经网络 (RNN) 和卷积神经网络 (CNN) 的设计, 完全依赖于自注意力机制 (self-attention mechanism) 来处理输入数据。

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机器翻译-Transformer

Transformer和机器翻译
Transformer and Machine Translation

```
class Transformer(nn.Module):
    def __init__(self, input_dim, output_dim, d_model, nhead,
                 num_encoder_layers, num_decoder_layers, dim_feedforward, dropout):
        super().__init__()
        self.embedding = nn.Embedding(input_dim, d_model)
        self.d_model = d_model
        self.transformer = nn.Transformer(d_model, nhead, num_encoder_layers,
                                          num_decoder_layers, dim_feedforward, dropout, batch_first=False)
        self.fc_out = nn.Linear(d_model, output_dim)
        self.dropout = nn.Dropout(dropout)
```

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机器翻译-Transformer

Transformer和机器翻译
Transformer and Machine Translation

```
def _generate_positional_encoding(self, seq_len):
    position = torch.arange(seq_len).unsqueeze(1)
    div_term = torch.exp(torch.arange(0, self.d_model, 2, dtype =
torch.float) * (- math.log(10000.0) / self.d_model))
    pe = torch.zeros(seq_len, self.d_model)
    pe[:, 0::2] = torch.sin(position * div_term)
    pe[:, 1::2] = torch.cos(position * div_term)
    return pe.unsqueeze(1) # [1, seq_len, d_model]
```

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机器翻译-Transformer

Transformer和机器翻译
Transformer and Machine Translation

```
def forward(self, src, trg, trg_mask = None, padding_mask=None):
    src_seq_length, N = src.shape
    trg_seq_length, N = trg.shape
    # 动态生成位置编码
    src_pos = self._generate_positional_encoding(src_seq_length).to(src.device)
    trg_pos = self._generate_positional_encoding(trg_seq_length).to(trg.device)
    # 将源位置编码的形状以匹配输入
    src_pos = src_pos.expand(-1, N, -1) # [1, seq_len, d_model] -> [seq_len, batch_size, d_model]
    trg_pos = trg_pos.expand(-1, N, -1) # [1, seq_len, d_model] -> [seq_len, batch_size, d_model]
    src = self.dropout(self.embedding(src) + src_pos)
    trg = self.dropout(self.embedding(trg) + trg_pos)
    if (trg_mask == None):
        output = self.transformer(src, trg)
    else:
        output = self.transformer(src, trg, trg_mask=trg_mask, tgt_key_padding_mask=padding_mask)
    prediction = self.fc_out(output)
    return prediction
```

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机器翻译-Transformer

Transformer和机器翻译
Transformer and Machine Translation

```
INPUT_DIM = len(chinese_vocab)
OUTPUT_DIM = len(english_vocab)
D_MODEL = 32
NHEAD = 2
NUM_ENCODER_LAYERS = 2
NUM_DECODER_LAYERS = 2
DIM_FEEDFORWARD = 32
DROPOUT = 0.05

model = Transformer(INPUT_DIM, OUTPUT_DIM, D_MODEL, NHEAD,
NUM_ENCODER_LAYERS, NUM_DECODER_LAYERS, DIM_FEEDFORWARD, DROPOUT)
criterion = nn.CrossEntropyLoss(ignore_index=0)
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
```

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机器翻译-Transformer

Transformer和机器翻译
Transformer and Machine Translation

```
for epoch in range(200):
    for i, (src, trg) in enumerate(data_loader):
        # 生成掩码
        trg_mask = nn.Transformer.generate_square_subsequent_mask(trg.size(0)-1).bool()
        padding_mask = (trg[1:,] == 0).transpose(0, 1) # (N, T)
        # 去掉最后一个token:trg[-1,]
        output = model(src, trg[:-1,], trg_mask=trg_mask, padding_mask=padding_mask)
        #print("output.shape:", output.shape)
        # 计算loss需要去掉第一个 token (BOS)
        loss = criterion(output.view(-1, OUTPUT_DIM), trg[1:,].view(-1))
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()
    if epoch % 100 == 99:
        print(f'Epoch {epoch + 1}, Loss: {loss.item()}')
```

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机器翻译评测

Transformer和机器翻译
Transformer and Machine Translation

机器翻译的评测是评估机器翻译系统输出质量的过程。这一过程对于改进现有系统、选择最佳模型以及了解翻译性能至关重要。以下是几种常见的机器翻译评测方法：

自动评测指标

BLEU (Bilingual Evaluation Understudy): 这是最常用的一种评分指标，通过比较翻译结果与参考译文之间n-gram重叠程度来评估翻译质量。BLEU分数越高，表示翻译结果与参考译文越接近。

ROUGE (Recall-Oriented Understudy for Gisting Evaluation): 主要用于评估摘要质量，但在某些情况下也可用于机器翻译的评测。

METEOR (Metric for Evaluation of Translation with Explicit ORdering): 综合考虑了词序和同义词等因素，通常被认为比BLEU更具可读性和准确性。

TER (Translation Error Rate): 计算翻译过程中所需进行的最小编辑操作次数（如插入、删除、替换）以匹配参考译文。

ChrF++: 类似于n-gram重叠，但使用字符n-grams而不是单词n-grams，被认为更能反映人类感知的翻译质量。

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机器翻译评测

Transformer和机器翻译
Transformer and Machine Translation

人工评测

人工评测通常由专家或非专家参与者完成，可以提供更细致入微的反馈，但成本较高且耗时较长。常见的人工评测方法包括：

- **直接评估:** 直接让评审者根据翻译结果的质量打分。
- **错误分类:** 通过详细记录翻译中的错误类型（如语法错误、漏译、误译等），来评估翻译系统的具体不足之处。
- **排序任务:** 让评审者根据翻译质量对多个翻译结果进行排序。
- **对比测试:** 将机器翻译结果与人类翻译结果进行对比，评估机器翻译的优劣。

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